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Subject - AM 102 (Mathematics - II)
Date - 03rd February, 2026

Assignment - I

① a)

$$A = \begin{pmatrix} 0 & -1 & 5 \\ 2 & 4 & -6 \\ 1 & 1 & 5 \end{pmatrix}$$

Using Row Elementary Operations to convert in Echelon form.

i) $R_3 \to 2R_3 - R_2 \Rightarrow A = \begin{pmatrix} 0 & -1 & 5 \\ 2 & 4 & -6 \\ 0 & -2 & 16 \end{pmatrix}$

ii) $R_1 \leftrightarrow R_2 \Rightarrow A = \begin{pmatrix} 2 & 4 & -6 \\ 0 & -1 & 5 \\ 0 & -2 & 16 \end{pmatrix}$

iii) $R_3 \leftrightarrow R_3 - 2R_2 \Rightarrow A = \begin{pmatrix} 2 & 4 & -6 \\ 0 & -1 & 5 \\ 0 & 0 & 6 \end{pmatrix}$

Since all rows are unique and none is a Zero Vector.

Hence, $\boxed{\text{Rank of Matrix } A = \text{No. of rows in } A = 3}$

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ii.)

$$A = \begin{pmatrix} 5 & 3 & 0 \\ 1 & 2 & -4 \\ -2 & -4 & 8 \end{pmatrix}$$

Using Row Elementary Operations to

a) $R_3 \to R_3 + 2R_2 \begin{pmatrix} 5 & 3 & 0 \\ 1 & 2 & -4 \\ 0 & 0 & 0 \end{pmatrix}$

Then using $R_2 \to 5R_2 - R_1$

$\Rightarrow B = \begin{pmatrix} 5 & 3 & 0 \\ 0 & 7 & -20 \\ 0 & 0 & 0 \end{pmatrix}$ Now B is in its Row Echelon Form and hence R_3 is Zero Vector and R_1 and R_2 are distinct.

$\therefore \boxed{\text{Rank of } B = 3 - 1 = 2}$

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iii) Given,

$$C = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 3 & 6 & 3 & -7 \end{pmatrix}$$

Using Elementary Operations,

a) $R_3 \to R_3 - 3R_1$ $\quad C = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 0 & 0 & 6 & -16 \end{pmatrix}$

Then, using

b) $R_2 \to R_2 - 2R_1$ $\quad C = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 6 & -16 \end{pmatrix}$

We can observe, $R_3 = 2R_2$

Therefore, $\boxed{\text{Rank of } C = 2}$

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(2) i) Given, $X = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{pmatrix}$

a) Perform $R_2 \to R_2 - 2R_1$ & $R_3 \to R_3 - 3R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$$

b) Now, $R_3 \to R_3 - R_2$

c) Multiply R_2 by -1: $R_2 \to -R_2$

Final echelon form: $X = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.

Rank of X

Number of non-zero rows = 2

$\boxed{\text{Rank}(X) = 2}$

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ii) $Y = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{pmatrix}$

a) Perform, $R_1 \to R_2$

$Y = \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 1 & 1 & 3 \end{pmatrix}$

b) Then, $R_3 \to R_3 - 3R_1$

$Y = \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & -3 \end{pmatrix}$

c) At last, $R_3 \to R_3 + 5R_2$

$Y = \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{pmatrix}$

All rows are unique, $\boxed{\text{Rank} = 3}$

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iii. Given, $Z = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 3 & 4 & 5 & 1 \\ 2 & 3 & 4 & 0 \end{pmatrix}$

a) Perform $R_3 \to R_3 - 2R_1$ & $R_2 \to R_2 - 3R_1$

$$Z = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -2 \end{pmatrix}$$

Then $R_3 \to R_3 - R_2$ & $R_2 \to -R_2$

$$Z = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & +1 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\boxed{\text{Rank}(Z) = 3}$$

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(3) Given, $A = \begin{pmatrix} 1 & 2 & 6 \\ 2 & 5 & 15 \\ 6 & 15 & 46 \end{pmatrix}$

Using Gauss - Jordan Method,

a) Form Augumented matrix:

$$[A|I] : \left[\begin{array}{ccc|ccc} 1 & 2 & 6 & 1 & 0 & 0 \\ 2 & 5 & 15 & 0 & 1 & 0 \\ 6 & 15 & 46 & 0 & 0 & 1 \end{array}\right]$$

b) $R_3 \to R_3 - 3R_2$ $\left[\begin{array}{ccc|ccc} 1 & 2 & 6 & 1 & 0 & 0 \\ 2 & 5 & 15 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -3 & 1 \end{array}\right]$

c) $R_2 \to R_2 - 2R_1$ $\left[\begin{array}{ccc|ccc} 1 & 2 & 6 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & -3 & 1 \end{array}\right]$

d) $R_2 \to R_2 - 3R_3$ & then $R_1 \to R_1 - 6R_3$ and $R_1 \to R_1 - 2R_2$

$$A^{-1} = \begin{bmatrix} 5 & 2 & 0 \\ -2 & 10 & -3 \\ 0 & -3 & 1 \end{bmatrix}$$

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(4) $2x + y + z = 5$ Here,
i) $x + y + z = 4$
$x - y + 2z = 1$

$$[A|B] = \left(\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 1 & 1 & 1 & 4 \\ 1 & -1 & 2 & 1 \end{array}\right)$$

And, $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{pmatrix}$

a) Perform $R_2 \to R_2 - \frac{1}{2}R_1$

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & -1 & 2 \end{pmatrix}$$

b) Perform, $R_3 \to R_3 - \frac{1}{2}R_1$

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{3}{2} & \frac{3}{2} \end{pmatrix}$$

c) Then, $R_3 \to R_3 + 3R_2$

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 3 \end{pmatrix}$$

$\boxed{\text{Rank}(A) = 3}$

Applying same operations,

$$[A|B] = \left(\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1.5 \\ 0 & 0 & 3 & 3 \end{array}\right)$$

$\boxed{\text{Rank of } [A|B] = 3}$

System is consistent and has unique solution.

i) $3z = 1$ $\boxed{z = 1}$

ii) $y/2 + z/2 = 3/2$ $\boxed{y = 2}$

iii) $2x + y + z = 5$ $\boxed{x = 1}$

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(4) ii. Given, $x + y + z = 6$

$x + 2y + 3z = 14$

$x + 4y + 7z = 30$

Here, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 7 \end{bmatrix}$ and $[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 1 & 4 & 7 & 30 \end{array}\right]$

a) Perform, $R_3 \to R_3 - R_2$ in A

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 2 & 4 \end{bmatrix}$$

b) Then, $R_2 \to R_2 - R_3$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & 2 & 4 \end{bmatrix}$$

c) After that, $R_2 \to R_2 - R_1$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 2 & 4 \end{bmatrix}$$

Here, we can observe $R_3 = -2R_2$

Therefore, $\text{Rank}(A) = 2$

After the same operations,

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

So, $y + 2z = 8$ and $x + y + z = 6$

Let, $z = K$ (free variable).

Ans: $x = K - 2$; $y = 8 - 2K$ & $z = K$

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(5) Given,

$$x - 4y + 7z = 14$$
$$3x + 8y - 2z = 13$$
$$7x - 8y + 26z = 5$$

Here, $A = \begin{bmatrix} 1 & -4 & 7 \\ 3 & 8 & -2 \\ 7 & -8 & 26 \end{bmatrix}$

a) Firstly perform, $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 7R_1$

$$A = \begin{bmatrix} 1 & -4 & 7 \\ 0 & 20 & -23 \\ 0 & 20 & -23 \end{bmatrix}$$

b) Then, $R_3 \to R_3 - R_2$

$$A = \begin{bmatrix} 1 & -4 & 7 \\ 0 & 20 & -23 \\ 0 & 0 & 0 \end{bmatrix}$$

After performing similar operations,

$$[A|B] = \left[\begin{array}{ccc|c} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 0 & 0 & -64 \end{array}\right]$$

$$\text{Rank}[A] \neq \text{Rank}[A|B]$$
$$2 \neq 3$$

Also, $0.x + 0.y + 0.z = 0$

And, $0 \neq -64$

This System is inconsistent.

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⑥ Given,

$$x + 2y - 3z = -2$$
$$3x - y - 2z = 1$$
$$2x + 3y - 5z = K$$

Here, $A = \begin{bmatrix} 1 & 2 & -3 \\ 3 & -1 & -2 \\ 2 & 3 & -5 \end{bmatrix}$ and $[A|B] = \left[\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 3 & -1 & -2 & 1 \\ 2 & 3 & -5 & K \end{array}\right]$

a) Perform $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 2R_1$

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -7 & 7 \\ 0 & -1 & 1 \end{bmatrix}$$

b) Then, $R_3 \to R_3 - \frac{1}{7}R_2$

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -7 & 7 \\ 0 & 0 & 0 \end{bmatrix}$$

Apply same operations, we get

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & -7 & 7 & 7 \\ 0 & 0 & 0 & K+3 \end{array}\right]$$

Rank of $[A]$ = Rank of $[A|B]$ = $2 < 3$

Ans: $\boxed{K = -3}$

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⑦ Given,

$$x + y + z = 7$$
$$x + 2y + 3z = 18$$
$$y + Kz = 6$$

Here, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 1 & K \end{bmatrix}$ and $[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 1 & 2 & 3 & 18 \\ 0 & 1 & K & 6 \end{array}\right]$

a) $R_2 \to R_2 - R_1$ $\quad A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & K \end{bmatrix}$

b) $R_3 \to R_3 - R_2$ $\quad A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & K-2 \end{bmatrix}$

Applying same operations,

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 0 & 1 & 2 & 11 \\ 0 & 0 & K-2 & -5 \end{array}\right],$$

Rank of $[A]$ = Rank of $[A|B] = 2 > 3$

For an inconsistent linear equations system, $\boxed{K=2}$

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(8) Given,

$$x + ay + z = 3$$
$$x + 2y + 2z = b$$
$$x + 5y + 3z = 9$$

Here, $A = \begin{bmatrix} 1 & a & 1 \\ 1 & 2 & 2 \\ 1 & 5 & 3 \end{bmatrix}$ and $[A|B] = \left[\begin{array}{ccc|c} 1 & a & 1 & 3 \\ 1 & 2 & 2 & b \\ 1 & 5 & 3 & 9 \end{array}\right]$

a) $\cancel{R_3 \to R_3 - \tfrac{5}{2}R_2 \text{ and } R_2 \to R_2 - R_1}$

$\cancel{A = \begin{bmatrix} 1 & a & 1 \\ 0 & 2-a & 1 \\ 0 & 5-a & 2 \end{bmatrix}}$

$|A| = 1[2.3 - 5.2]$
$\quad -a[1.3 - 2.1]$
$\quad +1[5.1 - 2.1]$

$|A| = -4 - a + 3 = -1 - a$

Compute $|A| = -(a+1)$

b) When $a \neq -1$

Then $|A| \neq 0 \Rightarrow \text{Rank}(A) = 3$

System is consistent and has a UNIQUE solution.

c) $|A| = 0 \Rightarrow \text{Rank} < 3$

For consistency $[b = ?]$

Put $a = -1$

$$[A|B] = \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 3 & 1 & b-3 \\ 0 & 6 & 2 & 6 \end{array}\right]$$

$R_3 \to R_3 - 2R_2$

$$[A|B] = \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 3 & 1 & b-3 \\ 0 & 0 & 0 & 12-2b \end{array}\right]$$

$$12 - 2b = 0$$

$$\boxed{b = 6}$$

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(9) i) Given, $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$

a) Find Eigenvalues, $|A - \lambda I| = 0$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{bmatrix} \Rightarrow |A - \lambda I| = (2-\lambda)[(2-\lambda)^2 - 1] - (-1)[(-1)(2-\lambda)]$$

$$\Rightarrow |A - \lambda I| = (2-\lambda)[\lambda^2 + 4 - 4\lambda - 1] - 2 + \lambda$$

$$= (2-\lambda)[\lambda^2 - 4\lambda + 2]$$

$$\lambda = 2$$

b) $A - 2I = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

$(A - 2I)x = 0$; $V_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$$\left[\frac{4 \pm \sqrt{16 - 4\cdot1\cdot2}}{2}\right] \to 2 \pm \sqrt{2} \qquad \sqrt{4\cdot2},\ \sqrt{8} = 2\sqrt{2}$$

c) $\lambda = 2 + \sqrt{2}$

$$[A - (2+\sqrt{I})I]x = \begin{bmatrix} -\sqrt{2} & -1 & 0 \\ -1 & -\sqrt{2} & -1 \\ 0 & -1 & -\sqrt{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -\sqrt{2}x_1 - x_2 \\ -x_1 - \sqrt{2}x_2 - x_3 \\ -x_2 - \sqrt{2}x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_1 = x_3 = \sqrt{2} + 1$

And, $x_2 = 1$

d) $\lambda = 2 - \sqrt{2}$

$A - \lambda I \Rightarrow [A - (2-\sqrt{2})I]x = 0$

$$\left[2-\sqrt{2}, \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}\right]$$

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9. ii) $B = \begin{pmatrix} 10 & -2 & 4 \\ -20 & 4 & -10 \\ -30 & 6 & -13 \end{pmatrix}$

$|B - \lambda I| = 0 \Rightarrow$ $B - \lambda I = \begin{pmatrix} 10-\lambda & -2 & 4 \\ -20 & 4-\lambda & -10 \\ -30 & 6 & -13-\lambda \end{pmatrix}$

$\lambda^3 - \lambda^2 - 2\lambda = 0$

$\lambda(\lambda^2 - \lambda - 2) = 0$

$\lambda(\lambda - 2)(\lambda + 1) = 0$

c) At $\lambda = 0$, $|A - \lambda I|x = |A|x = 0$

$$\begin{bmatrix} 10 & -2 & 4 \\ -20 & 4 & -10 \\ -30 & 6 & -13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} -10x_1 - 2x_2 + 4x_3 \\ -20x_1 + 4x_2 - 10x_3 \\ -30x_1 + 6x_2 - 13x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Ans: $\left[0, \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix}\right], \left[-1, \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}\right], \left[2, \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}\right]$

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(10) To prove: A^T has eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.
Given: $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of the matrix A.
i) Proof: Characteristic equation:

$$|A - \lambda I| = 0$$

Conside A^T, then: $|A^T - \lambda I| = 0$

But, $A^T - \lambda I = (A - \lambda I)^T$

Also, $|M^T| = |M|$

Therefore, $|A - \lambda I| = |A^T - \lambda I|$

Hence Proved

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To prove: A^{-1} has eigenvalues $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots \frac{1}{\lambda_n}$, $\lambda_i \neq 0$ for $i = 1, 2, \dots, n$.

Proof: If $\underline{|A - \lambda I| = 0}$; Then, $|A^{-1} - \frac{1}{\lambda} \cdot I| = 0$

$$Ax = \lambda x$$

Multiply both sides by A^{-1},

$$A^{-1}(Ax) = A^{-1}(\lambda x)$$
$$x = \lambda A^{-1}x$$
$$A^{-1}x = \frac{1}{\lambda}x$$

Hence Proved.

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iii) $Ax = \lambda x \quad \ldots (1)$

$$A^2x = A(Ax) = A(\lambda x) = \lambda(Ax) = \lambda^2 x = \lambda(\lambda x)$$

So, $A^2x = \lambda^2 x$

Similarly, $A^3x = \lambda^3 x$

and in general, $A^k x = \lambda^k x$

λ^k is an eigenvalue of A^k.

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11. A system has non-trivial solution, if $|A| = 0$

$$A = \begin{bmatrix} 1 & \cos\gamma & \cos\beta \\ \cos\gamma & 1 & \cos\alpha \\ \cos\beta & \cos\alpha & 1 \end{bmatrix}$$

$$|A| = 1 - \cos^2\alpha - \cos\gamma(\cos\gamma - \cos\alpha.\cos\beta) + \cos\beta(\cos\gamma.\cos\alpha - \cos\beta)$$

$$|A| = 1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma + 2\cos\alpha.\cos\beta.\cos\gamma$$

Given, $\underline{\alpha + \beta + \gamma = 0}$

(1).... $|A| = \sin^2\alpha + \sin^2\beta + \sin^2\gamma - 2\cos\alpha.\cos\beta.\cos\gamma$ $[\because 1 - \cos^2\theta = \sin^2\theta]$

$$\cos(\alpha+\beta+\gamma) = \cos 0 = 1$$

$\Rightarrow \cos\alpha.\cos\beta.\cos\gamma - \cos\alpha.\sin\beta.\sin\gamma - \sin\alpha.\cos\beta.\sin\gamma - \sin\alpha.\sin\beta.\cos\gamma = 1$

Rearrange give same eq^n (1).

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12) $\begin{bmatrix} 4 & 4 & -3 & 1 \\ 1 & 1 & -1 & 0 \\ K & 2 & 2 & -2 \\ 9 & 9 & K & 3 \end{bmatrix}$ $\begin{matrix} R_1 \to R_1 - 4R_2 \\ R_4 \to R_4 - 9R_2 \end{matrix}$ $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & -1 & 0 \\ K & 2 & 2 & -2 \\ 0 & 0 & K+9 & 3 \end{bmatrix}$

Since this is a 4x4 matrix, its maximum rank is 4.
The rank will be less than 4, $|A| = 0$.

$$|A| = (-1)^{1+3}(1)M_{13} + (-1)^{1+4}(1)M_{14}$$

$$M_{13} = \begin{vmatrix} 1 & 1 & 0 \\ K & 2 & -2 \\ 0 & 0 & 3 \end{vmatrix} = 3\begin{vmatrix} 1 & 1 \\ K & 2 \end{vmatrix} = 3(2-K)$$

$$M_{14} = \begin{vmatrix} 1 & 1 & -1 \\ K & 2 & 2 \\ 0 & 0 & K+9 \end{vmatrix} = K+9\begin{vmatrix} 1 & 1 \\ K & 2 \end{vmatrix} = K+9(2-K)$$

$$|A| = 6 - K - (2K + 18 - K^2 - 9K)$$

$$|A| = K^2 + 4K - 12$$

$$\therefore (K-2)(K+6) = 0$$

$$K = 2 \text{ or } K = -6$$

On checking minors, the rank does not drop to 2.

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13. $4x - 6y = -11$
$-3x + 8y = 10$
~~$4y + 3z = 8$~~

$A = \begin{bmatrix} 4 & -6 \\ -3 & 8 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \end{bmatrix}$; $B = \begin{bmatrix} -11 \\ 10 \end{bmatrix}$

a) $R_1 \to R_1 + R_2 \Rightarrow [A|B] = \left[\begin{array}{cc|c} 1 & 2 & -1 \\ -3 & 8 & 10 \end{array}\right]$

b) $R_2 \to R_2 + 3R_1 \Rightarrow [A|B] = \left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 14 & 7 \end{array}\right]$

c) $R_2 \to \frac{1}{7}R_2 \Rightarrow [A|B] = \left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 2 & 1 \end{array}\right]$

d) $R_1 \to R_1 - R_2 \Rightarrow [A|B] = \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 2 & 1 \end{array}\right]$

e) $R_2 \to \frac{1}{2}R_2 \Rightarrow [A|B] = \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 0.5 \end{array}\right]$

So we got $C = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ as the unique solution of given system of eq^n by Gauss Elimination Method.

$\therefore \boxed{x = 2, \ y = 0.5}$

ii. $4y + 3z = 8$ Given system can be represented

$2x - z = 2$ $\quad AX = B$

$3x + 2y = 5$ where $A = \begin{bmatrix} 0 & 4 & 3 \\ 2 & 0 & -1 \\ 3 & 2 & 0 \end{bmatrix}$

$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $B = \begin{bmatrix} 8 \\ 2 \\ 5 \end{bmatrix}$

Applying Elementary Row transformation on $[A|B]$

$$[A|B] = \left[\begin{array}{ccc|c} 0 & 4 & 3 & 8 \\ 2 & 0 & -1 & 2 \\ 3 & 2 & 0 & 5 \end{array}\right]$$

a) $R_1 \leftrightarrow R_2$ $\quad [A|B] = \left[\begin{array}{ccc|c} 2 & 0 & -1 & 2 \\ 0 & 4 & 3 & 8 \\ 3 & 2 & 0 & 5 \end{array}\right]$

b) $R_3 \to 2R_3 - 3R_1$ $\quad [A|B] = \left[\begin{array}{ccc|c} 2 & 0 & -1 & 2 \\ 0 & 4 & 3 & 8 \\ 0 & 4 & 3 & 4 \end{array}\right]$

c) $R_3 \to R_3 - R_2$ $\quad [A|B] = \left[\begin{array}{ccc|c} 2 & 0 & -1 & 2 \\ 0 & 4 & 3 & 8 \\ 0 & 0 & 0 & -4 \end{array}\right]$

$\Rightarrow$ As $[A|B]$ is in Row Echelon Form

$\Rightarrow$ Rank of $[A|B] = 3$ but Rank of $A = 2$

$\therefore$ No solution and hence system is inconsistent.

(E) 14. i. $\begin{bmatrix} 1.80 & -2.32 \\ -0.25 & 0.60 \end{bmatrix} = A = \begin{bmatrix} 9/5 & 58/25 \\ -1/4 & 3/5 \end{bmatrix}$

$[A|I] = \left[\begin{array}{cc|cc} 1.8 & -2.32 & 1 & 0 \\ -0.25 & 0.60 & 0 & 1 \end{array}\right] \Rightarrow$ ~~$R_2 \to R_2 + \frac{5}{36}R_1$~~ $\cancel{\begin{bmatrix} 1.8 & -2.32 \\ 0 & 83/90 \end{bmatrix}}$

$R_1 \to R_1 + \left(\frac{0.8}{0.25}\right)R_2$ $\quad [A|I] = \left[\begin{array}{cc|cc} 1 & -0.4 & 1 & 3.2 \\ -0.25 & 0.6 & 0 & 1 \end{array}\right]$

ii. Applying $R_2 \to 4R_2 + R_1 \Rightarrow [A|I] = \left[\begin{array}{cc|cc} 1 & -0.4 & 1 & 3.2 \\ 0 & 2 & 1 & 7.2 \end{array}\right]$

$R_1 \to 5R_1 + R_2 \Rightarrow [A|I] = \left[\begin{array}{cc|cc} 5 & 0 & 6 & 23.2 \\ 0 & 2 & 1 & 7.2 \end{array}\right]$

$R_1 \to \frac{1}{5}R_1$

$R_2 \to \frac{1}{2}R_2$ $\Rightarrow [A|I] = \left[\begin{array}{cc|cc} 1 & 0 & 1.2 & 4.64 \\ 0 & 1 & 0.5 & 3.6 \end{array}\right]$

$$A^{-1} = \begin{bmatrix} 1.2 & 4.64 \\ 0.5 & 3.6 \end{bmatrix}$$

iii. $A = \begin{bmatrix} 0.3 & -0.1 & 0.5 \\ 2 & 6 & 4 \\ 5 & 0 & 9 \end{bmatrix} \Rightarrow$ Applying $R_1 \to 10R_1$; $R_1 - R_2$ & $R_2 \to 5R_2 - 2R_3$.
Then, $R_3 \to R_3 - 5R_1$; $R_1 \to 5R_1 + R_3$ & $R_3 \to 6R_3 - 7R_2$

In end, $R_1 \to 10R_1 - 4R_3$ & $R_1 \to \frac{1}{50}R_1$; $R_2 \to \frac{1}{150}R_2$; $R_3 \to \frac{1}{10}R_3$
$R_2 \to 5R_2 - R_3$

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 54 & 0.9 & -3.4 \\ 0 & 1 & 0 & 2 & 0.2 & -0.2 \\ 0 & 0 & 1 & -30 & -0.5 & 2 \end{array}\right]$$

$\therefore$ Inverse of $A = \begin{bmatrix} 54 & 0.9 & -3.4 \\ 2 & 0.2 & -0.2 \\ -30 & -0.5 & 2 \end{bmatrix}$

25|B03|036

(15) Given,

$$x + y + z = 1$$
$$x + 2y + 4z = \alpha$$
$$x + 4y + 10z = \alpha^2$$

Here,
$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{bmatrix} \text{ and } [A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & \alpha \\ 1 & 4 & 10 & \alpha^2 \end{array}\right]$$

ii. $R_3 \to R_3 - 3R_2$
$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

i. $R_2 \to R_2 - R_1$
$R_3 \to R_3 - R_1$
$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 3 & 9 \end{bmatrix}$$

Apply same operations,
$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & \alpha - 1 \\ 0 & 0 & 0 & \alpha^2 - 1 - 3(\alpha - 1) \end{array}\right]$$

For consistency; $\alpha^2 - 1 - 3\alpha + 3 = 0$

$$\alpha^2 - 3\alpha + 2 = (\alpha - 1)(\alpha - 2) = 0$$

Ans: $\alpha = 1$ and $\alpha = 2$

25|303|036

(16) Given,

$$x + y + z = 6$$
$$x + 2y + 3z = 10$$
$$x + 2y + \lambda z = \mu$$

Here,
$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \text{ and } [A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right]$$

a. $R_3 \to R_3 - R_2$ and $R_2 \to R_2 - R_1$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & \lambda-3 \end{bmatrix}$$

b) Apply same operations,

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right]$$

i. $\lambda - 3 = \mu - 10$

No solution, $\boxed{\lambda = 3, \mu \neq 10}$

ii. Unique solution, $\boxed{\lambda \neq 3}$

iii. Infinite solution, $\boxed{\lambda = 3 \text{ and } \mu = 10}$

25/B03/036

(17) Given, $B = 2A^2 - \frac{1}{2}A + 3I$

And, $A = \begin{bmatrix} 6 & 4 \\ 4 & 0 \end{bmatrix}$

Compute, $A^2 = A * A = \begin{bmatrix} 6 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} 6 & 4 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 52 & 24 \\ 24 & 16 \end{bmatrix}$

$$B = \begin{bmatrix} 104 & 48 \\ 48 & 32 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 104 & 46 \\ 46 & 35 \end{bmatrix}$$

$$|B - \lambda I| = \begin{vmatrix} 104-\lambda & 46 \\ 46 & 35-\lambda \end{vmatrix} = (104-\lambda)(35-\lambda) - 46(46) = 0$$

$|B - \lambda I| = 3640 - 104\lambda - 35\lambda + \lambda^2 - 2116$

$\lambda^2 - 139\lambda + 1524 = 0$

$(\lambda - 127)(\lambda - 12) = 0$

Eigenvalue of A,

$$|A-\lambda I| = \begin{vmatrix} 6-\lambda & 4 \\ 4 & -\lambda \end{vmatrix} = (6-\lambda)(-\lambda) - 16 = -6\lambda + \lambda^2 - 16 = (\lambda-8)(\lambda+2)$$

Eigenvector,

$$B - 127I = \begin{pmatrix} -23 & 46 \\ 46 & -92 \end{pmatrix} = \begin{pmatrix} 104-127 & 46 \\ 46 & 35-127 \end{pmatrix}$$

$$(B-127I)x = 0 \Rightarrow \begin{pmatrix} -23 & 46 \\ 46 & -92 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$\Rightarrow -23x + 46y = 0 \Rightarrow x = 2y$ - (i)

$\Rightarrow$ Take $y = 1 \Rightarrow x = 2$

Similarly, $B - 12I = 0$

Ans: $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

25|1363|036

(18) Given,

$$3x + 4y + 5z = a$$
$$4x + 5y + 6z = b$$
$$5x + 6y + 7z = c$$

Here,

$$[A|B] = \left[\begin{array}{ccc|c} 3 & 4 & 5 & a \\ 4 & 5 & 6 & b \\ 5 & 6 & 7 & c \end{array}\right]$$

$R_3 \to R_3 - R_2$
$R_2 \to R_2 - R_1$

$$[A|B] = \left[\begin{array}{ccc|c} 3 & 4 & 5 & a \\ 1 & 1 & 1 & b-a \\ 1 & 2 & 1 & c-b \end{array}\right]$$

$$[A|B] = \left[\begin{array}{ccc|c} 3 & 4 & 5 & a \\ 1 & 1 & 1 & b-a \\ 0 & 0 & 0 & c+a-2b \end{array}\right] \quad R_3 \to R_3 - R_2$$

Rank of $[A] = 2$

Rank of $[A|B] = 3$

If $c+a-2b \neq 0$; we will get No solution conditions.
If we want a solution (atleast 1) then
$$\Rightarrow \boxed{a+c=2b}$$

If $a=b=c=-1$
then, $\boxed{x+y+z=0}$
$$\boxed{y=1-2z}$$
$$\boxed{x=z-1}$$

$\therefore$ If $z=\alpha$, solution set $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha-1 \\ 1-2\alpha \\ \alpha \end{bmatrix}$

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